Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

____________show ip route ospf____ 2

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? __**Serial 0/0/1**__ 0/0/0 0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? ___129____ 2

Does R2 show up as an OSPF neighbor on R1? __Yes___ 2

Does R2 show up as an OSPF neighbor on R3? ___No____ 2

What does this information tell you?

___OSPF routing can be learn through other path, not only through direct connection. 1

____

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

______________R2(config)# router ospf 1____

_________R2(config-router)# no passive-interface s0/0/1_____________________________________ 2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ______**Serial 0/0/0** _ 0/0/1 0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

_____65,cost from r3 to r2 is 64 +1 = 65 2

____

Is R2 listed as an OSPF neighbor to R3? __Yes___ 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

______ modern infrastructure devices have links that are faster than default reference-bandwidth for OSPF， lead to a result that OSPF cannot identify the quality of the connection.______________________

____

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

_____To 192.168.3.0/24: 100mbs/128 =781 ,781+1=782____________________

_____To 192.168.23.0/30 :_781+64 =845____________________________

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

______1562,since the bandwidth become 128k, so 781+781=1562_________________________________

____

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

__Since ip ospf cost 1565,the cost is now higher than through R2.

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Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   ___If router ID change, OSPF need to restart and calculate the routing and shut down the network, To ensure the network stable, an assigned router ID can prevent this chaos.

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2. Why is the DR/BDR election process not a concern in this lab?

   ______Because it is a point to point network.

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3. Why would you want to set an OSPF interface to passive?

   ___Prevent sending routing data to unsecure LAN.

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4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   _________ network 10.0.0.0 0.255.255.255 area 0

   4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   _________RIP,since the metric is smaller.

   0